

FEASIBLE-TAIL TEST-TIME ADAPTATION FOR AUDIO DEEPPFAKE DETECTION UNDER PREVALENCE SHIFT

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ABSTRACT

Confident-tail test-time adaptation (TTA) pseudo-labels the extreme quantiles of a detector’s scores on unlabelled target audio and self-trains on them. The rule is *infeasible* whenever the deployment pool is class-skewed: a counting bound caps pseudo-label purity at $\min(1, \pi_c/q_c)$ for any detector, so a symmetric budget q needs $q \leq \min(\pi, 1-\pi)$. We propose **Feasible-Tail Adaptation (FTA)**, which sizes each tail to the class it is meant to contain, $q_c=2q\hat{\pi}_c$ from a label-free prior estimate, capped by a prevalence bound where no estimate can be trusted. FTA is identical to the symmetric rule on balanced pools and repairs it where it fails. On ASVspoof2021-DF (97.2% spoof) it lifts pseudo-real tail purity from 0.09 to 0.99, turns a 63.3% accuracy collapse back into 98.9%, and improves EER on 8 of 9 runs ($4.30 \rightarrow 3.81$); on In-the-Wild resampled to 90–97% fake it adds ~ 9 points of balanced accuracy.

Index Terms— audio deepfake detection, anti-spoofing, test-time adaptation, label shift, cross-corpus generalization.

1. INTRODUCTION

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1–3]. Deployed on audio from a different source, accuracy drops steeply, a well-documented generalization gap [5, 6, 23].

We start from a diagnostic measurement: a fine-tuned XLS-R detector on an unseen corpus keeps a high **ROC-AUC while its accuracy at a fixed threshold collapses**. It still *rank*s clips correctly, so its most-confident predictions on unlabelled target audio are trustworthy enough to adapt on.

Every confident-tail method takes the same step—label the extreme quantiles of the target scores and self-train on them—and that step carries an assumption that is invisible on the benchmarks the field uses and fatal on the pools it deploys to: that the target pool is roughly class-balanced. Take the top and bottom 30% of a 97%-spoof pool and the “confidently bona fide” bucket is mechanically made of low-scoring *spoof*. This is not hypothetical: it is what happens to

all three publicly released checkpoints we test on the official ASVspoof2021-DF protocol (Table 1), and to three of four modern TTA baselines.

Feasible-Tail Adaptation. The fix follows from asking how large a tail may be: a bucket of $q_c N$ clips cannot hold more than the $\pi_c N$ clips of class c that exist, so no detector can push its purity past $\min(1, \pi_c/q_c)$. FTA sizes each tail to its class, $q_c=2q\hat{\pi}_c$, from a label-free prior estimate, and where no estimate can be trusted caps q by a prevalence lower bound instead. At $\hat{\pi}=\frac{1}{2}$ it is *identical* to the symmetric rule: it generalizes the recipe it repairs, and inherits its behaviour on the balanced pools where that recipe was validated.

Contributions. (i) FTA: a prevalence-aware tail budget that is a strict generalization of the symmetric rule, needs no target label, and costs one scalar per class. (ii) The feasibility constraint it is built from, $q_c \leq \pi_c$ — exact, detector-independent, and confirmed by intervening on prevalence directly. (iii) An evaluation on released checkpoints at the benchmark’s own prevalence: damage prevention on DF, balanced-accuracy gains on skewed In-the-Wild pools, parity on balanced pools, operating-point repair over 369 third-party cells. (iv) The scope condition: the prior estimate degrades as the source’s operating point does ($r=0.997$), so FTA ships with an estimator-free cap and two label-free monitors, one recovering 13.9 EER points from a Tent collapse.

Related work. SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1–4, 21]. Train-time domain generalization aligns source domains [5] or runs gradient reversal over source and target jointly [15]; both act before deployment. In TTA, entropy minimisation (Tent) [7] is the reference recipe, and its instability prompted reliable-sample filtering (ETA/EATA) [8], sharpness-aware entropy with recovery (SAR) [9], and information maximisation with self-labelling (SHOT) [10] — we compare against all four. Closest to us is T²A [20], which also targets deepfake TTA and diagnoses entropy minimisation’s confirmation bias, replacing it with uncertainty-aware negative learning on image/video deepfakes; it repairs the *objective*, while we show the *sampling* carries a prevalence precondition no objective can remove. Adaptation under label shift is itself established

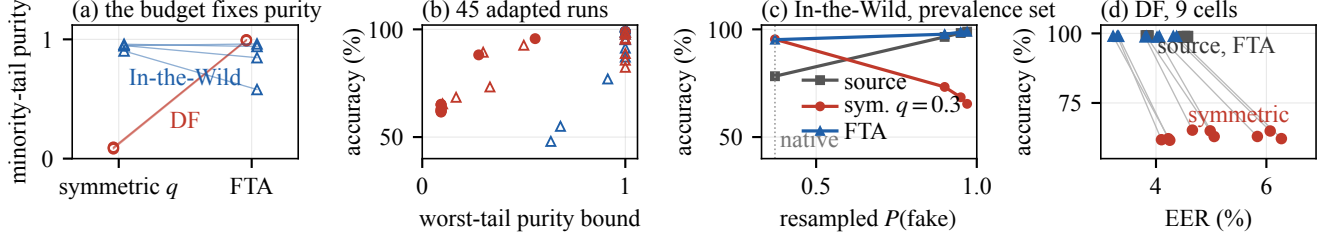


Fig. 1: What the budget does. (a) Realised minority-tail purity under the symmetric budget and under FTA, one marker per checkpoint: on DF (red) FTA lifts it from 0.09 to 0.99; on In-the-Wild (blue) the symmetric budget is already feasible. (b) Accuracy vs. the worst-tail bound of Eq. (1), every adapted run on DF (•) and In-the-Wild (△); red symmetric, blue FTA. (c) In-the-Wild resampled to a target $P(\text{fake})$, audio and weights fixed; the smaller budgets reported in the text lie between the two curves. (d) Official DF, nine runs.

— Label Shift Adapter [11] adapts from an estimated target prior, RLSbench [12] benchmarks relaxed label shift — but both take the estimate as given, whereas the bound below constrains any tail budget without one, and the estimate degrades exactly where it is needed (Sec. 4.6). The prior estimator is black-box shift estimation [13]; the proxy $\mathcal{A}$ -distance is from [17].

2. METHOD

Let x be a waveform with label $y \in \{0, 1\}$ (fake = 1) and let f_θ produce $s_\theta(x) = P_\theta(y=1 | x)$. We are given a source model, its labelled training corpus $\mathcal{S}$, and an *unlabelled* target pool $\mathcal{T}$. No target label is read at any point.

What breaks under corpus shift. AUC is threshold-free; accuracy at a fixed τ is not, and empirically AUC degrades far less: across 369 third-party cells the accuracy a model’s ranking supports but its shipped threshold does not deliver averages 7.29 points. The deployable object is the score *order*; the broken one is the operating point.

(a) How large may a tail be? The symmetric rule — our own earlier configuration, an instance of confidence-based pseudo-label selection [10, 19] — takes the bottom q and top q of the target scores as pseudo-real and pseudo-fake. Its ceiling needs no model at all. A bucket of $q_c N$ clips can contain at most the pool’s $\pi_c N$ class- c clips, so for any detector whatsoever

$$\text{purity}_c \leq \min(1, \pi_c / q_c). \quad (1)$$

This is counting, not an approximation; what is empirical is its *tightness*. Where prevalence binds, it is tight: on DF the realised tails sit within 0.0003 of the bound (Fig. 1a). Where it is slack, the ranking decides how much of the headroom is used. A symmetric budget therefore admits pure pseudo-labels only while

$$q \leq \min(\pi, 1 - \pi), \quad (2)$$

which In-the-Wild satisfies at $q=0.3$ (min = .372) and DF does not (min = .028): there no detector can exceed 9.3% pseudo-real purity, we measure 8.9%. The damage is one-sided — DF’s pseudo-fake tail is pure at every q — which is

what makes it repairable by budgeting rather than by a better model.

(b) Feasible-Tail Adaptation. FTA gives each class the budget its own prevalence can fill,

$$q_c = 2q \hat{\pi}_c, \quad (3)$$

so Eq. (1) is satisfied for *any* $q \leq \frac{1}{2}$ under a correct prior, and $q_c = q$ at $\hat{\pi} = \frac{1}{2}$: on a balanced pool FTA is the symmetric rule, byte for byte. With an estimate it needs

$$\hat{\pi}_c / \pi_c \leq 1 / (2q) \quad \text{for both classes}, \quad (4)$$

i.e. at $q=0.3$ the estimate may overstate either class by at most $1.67\times$ — satisfied on DF (1.03) and violated on the one cell where a source model arrives badly mis-calibrated (Sec. 4.6). A budget needs the prior only to within a factor, not exactly, which is why an estimator that is unidentifiable in general is still usable here. Because q is ours to choose and π is not, FTA also admits an *estimator-free* instantiation: cap q at a lower bound on $\min(\pi, 1 - \pi)$, buying safety with supervision ($q=0.02$ labels 800 of 20,000 clips against 12,000). We report both, and a fixed-confidence prior-free control.

(c) The prior, label-free. $\hat{\pi}$ comes from black-box shift estimation [13]: $\hat{p} = M^{-T} h$ with M the source confusion matrix and h the target prediction histogram; a score mixture and an oracle prior are controls [14].

(d) Objective and guards. With pseudo-labels $\tilde{y}$ from (b) on the confident set $\mathcal{C}$ and a stochastic channel perturbation $a(\cdot)$ (per-clip gain $g \sim \mathcal{U}(0.7, 1.3)$ plus Gaussian noise $\sigma=0.005$),

$$\min_{\{\gamma, \beta\}, \phi} \underbrace{\sum_{x \in \mathcal{C}} \text{CE}(f_\theta(x), \tilde{y})}_{\mathcal{L}_{\text{ST}}} + \lambda \underbrace{\|f_\theta(a(x)) - \text{sg}[f_\theta(x)]\|^2}_{\mathcal{L}_{\text{cons}}}, \quad (5)$$

optimising only the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned blocks and the head ϕ (16,706 parameters), $\lambda=0.3$. $\mathcal{L}_{\text{cons}}$ is inherited from the base recipe and is *not* part of the contribution: dropping it moves DF EER by 0.08 (3.98 vs. 4.06) and costs less balanced accuracy (86.58 vs. 82.34 on seed 2). FTA is the budget, with or without it. Two label-free statistics guard the run at deployment — rank agreement with the frozen source and the drift of the predicted positive rate from $\hat{\pi}$ (Sec. 4.5).

3. EXPERIMENTAL SETUP

Protocol A. The official ASVspoof2021-DF eval partition [22], scored against the official CM keys. Of its 533,928 eval trials we hold 400,435 (74.998%, with similar coverage of either class), so our numbers are pool numbers and we make no full-set or SOTA comparison. True spoof rate 97.22%. We adapt *released* checkpoints: DeepFense XLS-R 300M+ AASIST trained on ASVspoof2019-LA [28], three independent training seeds. Adaptation uses an unlabelled 20,000-clip subsample; we report EER over the pool we hold, an inductive number over trials *not* in it, and two further seeds on a fixed 50,000 subset: nine runs. Metrics come from a re-audit of the cached score arrays (interpolated ROC crossing); Protocol B keeps its per-fold EERs.

Protocol B. Leave-one-corpus-out over ASVspoof2019-LA [24], D2 (LibriSpeech vs. eight open TTS systems), In-the-Wild [23] and the Arabic set [26]; each source pool merges the other three with a 38-language MLAAD subset [25]; target audio never appears in any source pool, and six MLAAD languages are held out of every pool. All methods run on the same twelve folds. **Protocol C** is all 31,779 In-the-Wild clips at their native 37.2% fake rate, skewed the *opposite* way to DF.

Comparison points. Source-only; the median-threshold rule; Tent, IM-PL, ETA/EATA and SAR; AdaBN [16]; DANN [15], which sees the held-out target’s unlabeled audio at every step; an ASDG-style baseline [5]; and a target-supervised oracle. Every TTA arm adapts the same 16,706 parameters with the same lr, batch and pass count; none was tuned separately, since per-target tuning needs the labels the setting assumes absent. EATA’s redundancy filter and SAR’s recovery trigger are degenerate at $C=2$ and were re-derived at their relative positions ($e_0 = (0.2 / \ln 1000) \ln C$).

Implementation. 16 kHz audio, top-4-layer XLS-R fine-tuning for our own source models (Adam, $lr 2 \times 10^{-4}$, batch 32, 8 epochs). Adaptation: $lr 1 \times 10^{-4}$, $E=4$ passes, batch 32 (third-party crops matched to each checkpoint’s own training length). Code and manifests: [github.com/-RohamIzadidoost/feasible-tail-adaptation](https://github.com/RohamIzadidoost/feasible-tail-adaptation), with per-arm score dumps that reproduce every number here without a GPU.

4. RESULTS

4.1. FTA on the official DF protocol

Table 1 is the main result. At this pool’s prevalence the symmetric budget cannot work and does not: accuracy 98.92 $\rightarrow$ 63.33%, EER 4.30 $\rightarrow$ 4.87, a 35-point threshold deficit where there was none. FTA, changing one scalar per class, keeps the accuracy (98.89%) and improves the ranking (4.30 $\rightarrow$ 3.72). The mechanism is the one Eq. (1) predicts: BBSE returns $\hat{\pi}=0.972$ against a true 0.9722, the pseudo-real budget

shrinks 0.30 $\rightarrow$ 0.014, and that tail’s purity rises 0.089 $\rightarrow$ 0.993 (Fig. 1a).

Across the nine runs (three checkpoints $\times$ three adaptation runs; reused checkpoints and overlapping pools make these sensitivity checks, not independent trials) FTA beats the symmetric budget 9/9 and the source 8/9 (4.30 $\rightarrow$ 3.81 mean EER), 7/9 on disjoint trials (4.31 $\rightarrow$ 3.83). Accuracy separates the arms with no overlap (Fig. 1d). Two caveats: balanced accuracy falls on two checkpoints and rises on the third (78.35 $\rightarrow$ 82.55), a loss that tracks $\mathcal{L}_{\text{cons}}$ rather than the budget; and DF-240 gains nothing on disjoint trials, so the improvement is carried by the two checkpoints with the larger deficits.

The baselines fail through their own versions of the balance assumption: IM-PL’s diversity term is maximised at a *uniform* mean prediction (at $C=2$, a hard prior of $\frac{1}{2}$), and ETA’s filter admitted 99.8% of the high-entropy samples it exists to discard, reducing it to Tent — an entropy key separates confident from uncertain, not confident-and-right from confident-and-wrong. SAR alone avoids damage, by not moving the model.

4.2. At native prevalence, FTA inherits the recipe

In-the-Wild at its native 0.372 satisfies Eq. (2) at $q=0.3$, where FTA and the symmetric rule nearly coincide and both repair rather than destroy. DeepFense s42 arrives ranking at AUC 0.917 while correct on 39.1% of clips, fewer than a majority-class predictor; adaptation drives the three mis-calibrated checkpoints’ deficits from 22.9, 16.8 and 48.3 to 3.3, 0.1 and 8.3, and raises s42’s AUC to .947 — which no threshold move can do — with the disjoint half slightly better, so the ranking gain is not memorisation.

4.3. Validating the budget rule by intervention

Sweeping q on DF s2 with everything else fixed drives accuracy 98.96 $\rightarrow$ 62.96 as the bound falls 1.000 $\rightarrow$ 0.093 (Spearman +1.000; Fig. 1b). At $q=0.02$, FTA’s estimator-free instantiation here, adaptation is safe and improves on source (4.51 $\rightarrow$ 4.19) with no prior at all, but stays weaker than the estimated budget (4.06): shrinking q discards the majority tail too.

Inducing the failure where the recipe works. Re-sampling In-the-Wild — where $q=0.3$ is feasible — to $P(\text{fake}) \in \{0.90, 0.95, 0.97\}$ with audio and weights fixed makes accuracy track the bound (Spearman +0.966; Fig. 1c): the symmetric $q=0.3$ collapses (73.3/68.5/65.4%), $q=0.1$ degrades only where its own bound does (95.9/92.6/89.4) and $q=0.02$ is unharmed (97.8/98.6/98.6), while FTA holds 97.7/98.7/98.9% and gains ~ 9 points of *balanced* accuracy on the source (93.7/91.6/92.3 against 82.7/83.6/83.5): the failure is prevalence, not DF.

Table 1: Official ASVspoof2021-DF eval, the 400,435 of 533,928 eval trials we hold (97.22% spoof), mean over the three released XLS-R+AASIST checkpoints [28]. All adaptation is label-free. Metrics are recomputed from the cached score arrays with an interpolated ROC crossing; δ is the exact best-threshold accuracy minus the realised one (an oracle diagnostic), and balanced accuracy is given because accuracy at 97% prevalence is majority-dominated.

	EER %	AUC	acc@0.5	bal. acc	δ
Source (as released)	4.30	.993	98.92	84.55	0.17
+ median threshold [¶]	4.30	.993	56.88	77.81	42.21
+ BBSE threshold [¶]	4.30	.993	98.96	85.39	0.12
+ Tent [7]	14.06 [‡]	.908	98.28	69.12	0.74
+ ETA [8]	31.15 [‡]	.748	97.79	60.90	0.59
+ IM-PL [10] [†]	5.65	.980	77.77	88.37	20.06
+ SAR [9]	4.26 [§]	.993	98.83	81.93	0.26
+ TTA, symmetric $q=0.3$	4.87	.990	63.33	81.12	35.32
+ FTA (ours)	3.72	.994	98.89	80.85	0.23

[¶] frozen-source threshold rules, no weights updated: a label-shift correction from the same $\hat{\pi}$ recovers accuracy, the prior-blind median rule does not, and neither can move EER or AUC. [†] probability-space information maximisation with self-labelling, SHOT-inspired rather than feature-space SHOT. [‡] EER over a pool that is mostly one tied score (Tent saturates 98.07% of scores to exactly 1.0, ETA 97.71%); read AUC. [§] SAR neither damages nor adapts: 308 recovery resets in 2,500 steps. Single-checkpoint arms (seed 2): FTA without $\mathcal{L}_{\text{cons}}$ 3.98, fixed confidence 0.95 4.30, and the estimator-free cap $q=0.02$ 4.19 against a 4.51 source.

Table 2: Protocol B, leave-one-corpus-out: twelve folds (four targets $\times$ three seeds) on our own XLS-R source models. Mean per-fold EER % $\pm$ s.d., accuracy at $\tau=0.5$, and folds improved over source (Wilcoxon p).

	EER %	acc@0.5	>src (p)
Source	18.39 $\pm$ 11.7	73.45	—
+ Tent [7]	32.25 $\pm$ 15.3	67.57	3/12 (.012)
+ ETA [8]	29.54 $\pm$ 15.4	69.77	2/12 (.003)
+ EATA [8]	31.34 $\pm$ 16.2	67.96	4/12 (.027)
+ SAR [9]	18.39 $\pm$ 11.7	73.22	5/12 (.91)
+ IM-PL [10]	17.76 $\pm$ 12.1	81.22	10/12 (.003)
DANN [15]	20.72 $\pm$ 12.5	71.84	4/12 (.11)
ASDG-style [5]	20.42 $\pm$ 11.6	77.10	3/12 (.15)
+ FTA (ours)	17.44 $\pm$ 12.1	81.09	10/12 (.027)
Target-supervised oracle	6.51 $\pm$ 4.9	89.62	12/12

4.4. Balanced pools: FTA is the recipe it generalizes

Where Eq. (2) holds, $\hat{\pi} \approx \frac{1}{2}$ and FTA reduces to the symmetric rule, so this arm confirms the generalization costs nothing (Table 2). FTA improves 10/12 folds over source and beats every entropy-based arm, while IM-PL ties it (17.76 vs. 17.44, $p=0.47$): a hard uniform prior is free where Eq. (2) holds and destructive where it does not, so the two separate only in the untested regime. Train-time baselines adapt less (DANN 20.72, ASDG-style 20.42); AdaBN is the source itself (18.57).

4.5. Label-free guards

Checking Eq. (2) needs π , not identifiable (Sec. 4.6), so a deployer asks the weaker question — *did* this adaptation help? — answered by two statistics: rank agreement with the frozen source $\rho = \text{Spearman}(s_\theta, s_{\theta_0})$ and the drift of the predicted positive rate from $\hat{\pi}$. ρ catches ranking collapse (ETA 0.31,

Tent 0.44), drift catches operating-point collapse (0.37, 0.39, 0.23), and together they separate the five harmful arms from the three harmless ones for any $\tau \in [0.70, 0.80]$; as a leave-one-target-out stopping rule ρ recovers 13.9 EER points from Tent’s collapse (oracle 17.24), never firing on FTA.

4.6. The target prior is not identifiable

A label-free estimate cannot be trusted far, for structural reasons rather than any defect of BBSE: it observes only $\Pr[\hat{y}=1] = \pi \text{TPR}_\mathcal{T} + (1-\pi)\text{FPR}_\mathcal{T}$, one equation in three unknowns, and closes it by assuming source error rates carry over — what covariate shift breaks. An extreme prior and an extreme calibration failure are then *indistinguishable* (DF/s2 predicts 97.4% fake at a true 0.972, ITW/s42 98.1% at a true 0.372), and $|\hat{\pi} - \pi|$ tracks the source deficit at $r=0.997$ over ten cells: estimating the prior and detecting that the estimate is wrong are one problem. ITW/s42 is accordingly the one cell where the estimated budget trails its own symmetric special case (14.54 EER against 12.58; 11.59 with the true prior), which is what the estimator-free instantiation is for — Eq. (2) needs only a *lower bound* on $\min(\pi, 1-\pi)$, and $q=0.02$ was safe on every cell here.

4.7. Breadth, and the mechanism

Over 369 third-party cells — nine released checkpoints (five architectures, incl. AST [18] and ShiftySpeech’s SSL-AASIST [27]) on the 76 pairs with no training overlap — the effect is *re-calibration*: δ falls 7.29 $\rightarrow$ 1.96 (248/369, $p=2.2 \times 10^{-17}$) while a proxy $\mathcal{A}$ -distance and a domain-origin probe both *rise* (Arabic 1.545 $\rightarrow$ 1.606): repair, not invariance. Eq. (2) is necessary, not sufficient.

Compliance with Ethical Standards

This work used only publicly released datasets and model checkpoints under their respective licences; no new human-subject data were collected and no human or animal experiments were performed. The authors declare no conflict of interest.

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